XD Technology Consulting Ltd

SUPPORTING STUDENT SUCCESS AND RETENTION THROUGH PREDICTIVE MODELLING

DATA SCIENCE REPORT

Application of Supervised Machine Learning Models for Student Dropout Prediction

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Public portfolio version using anonymised and obfuscated data.

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2 Introduction

Student retention is critical for institutional performance and student success. High dropout rates can lead to revenue loss, reduced reputation, and lower student satisfaction. Early identification of at-risk students allows targeted intervention and improved outcomes.

This report analyses student data provided by a partner organisation to evaluate supervised learning techniques for predicting student dropout.

The objective is to determine how early reliable predictions can be made, which types of student data provide the strongest predictive signal, and how these insights can be used to improve retention and reduce financial loss.

The analysis compares three progressively richer datasets. It evaluates predictive performance at each stage, but does not establish the earliest reliable prediction point because feature availability has not been evaluated against a defined prediction date.

3 Methodology

The same modelling workflow was applied at each data stage.

Workflow

1. Data exploration and feature engineering.
2. Training supervised machine learning models using the current stage dataset **'the baseline'**
3. Evaluating **baseline model performance**, with the impact of newly-added features*
   *Stage 2/3 only
4. **Hyperparameter tuning** and comparison against the baseline model.
5. Comparing model performance across stages to **assess predictive performance.**

Each data stage is summarised in Figure 1.

Stages

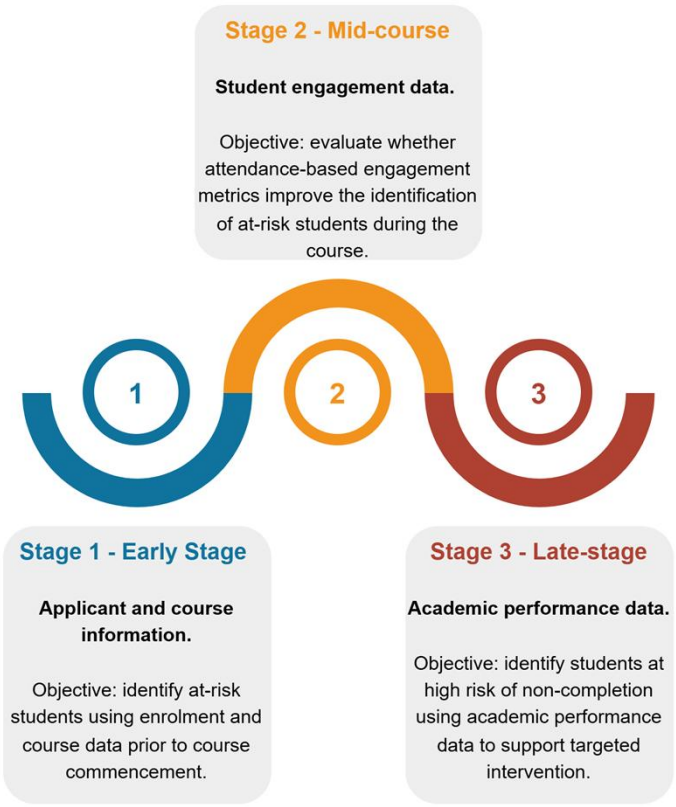


Figure 1 – Data Stages and their Business Objectives

3.1 Feature Engineering and Exploratory Data Analysis (EDA)

Feature engineering was performed to construct predictive variables aligned with the business objective of identifying student dropout risk at different stages of the course. Figure 1 summarises the data preparation and feature engineering process.

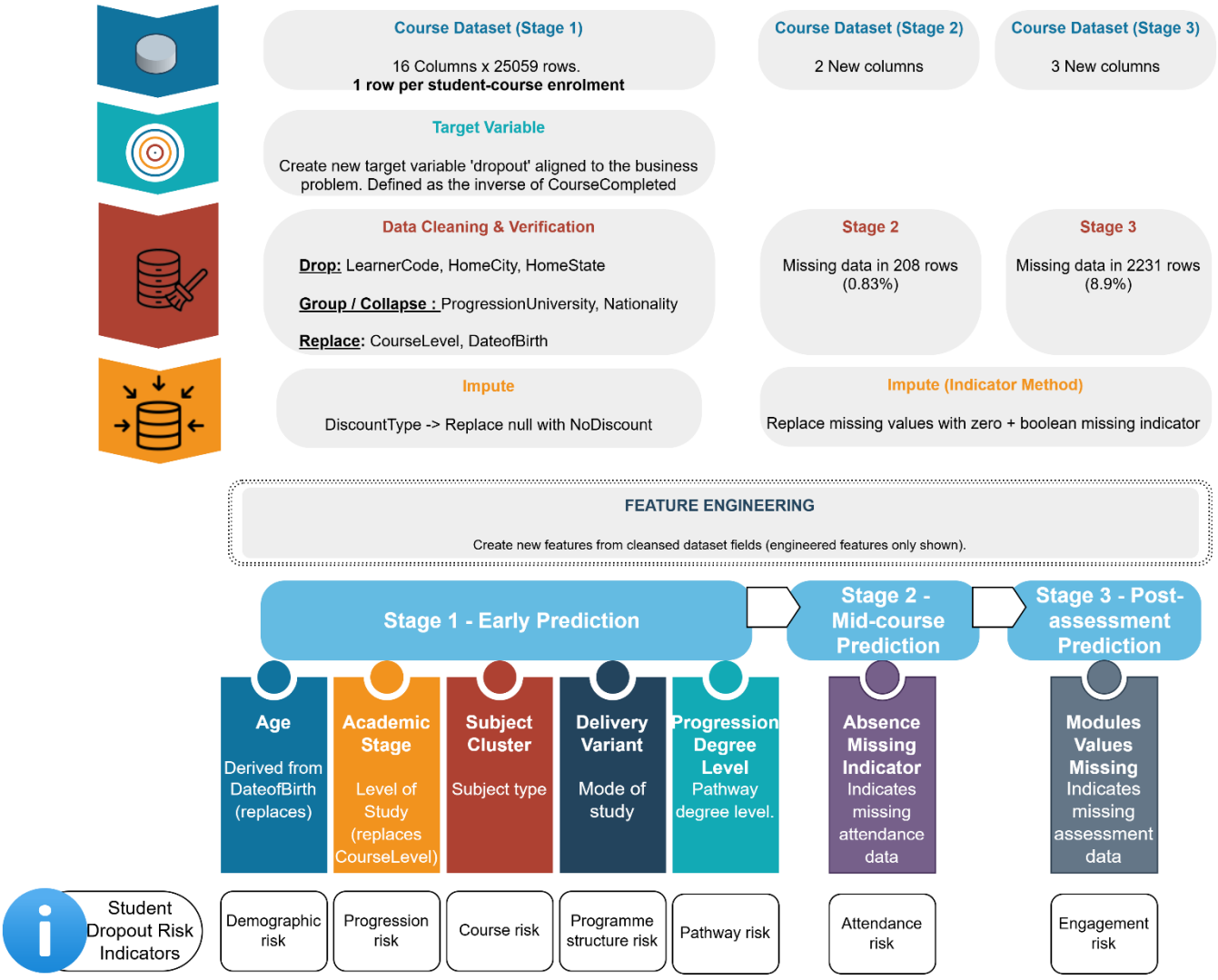


Figure 2 – Feature Engineering Process & Definitions

Descriptive statistics and visual inspections (via box-plots, histograms and Q-Q plots) were used to assess feature distributions and identify any outliers.

Categorical variables were plotted against the mean target value to show variation in predictive signal.

A full list of features can be found in Appendix A

3.2 Model Development Approach

Two model types were used: XGBoost and a feed-forward neural network. The model construction, tuning, and evaluation workflow is shown in Figure 3.

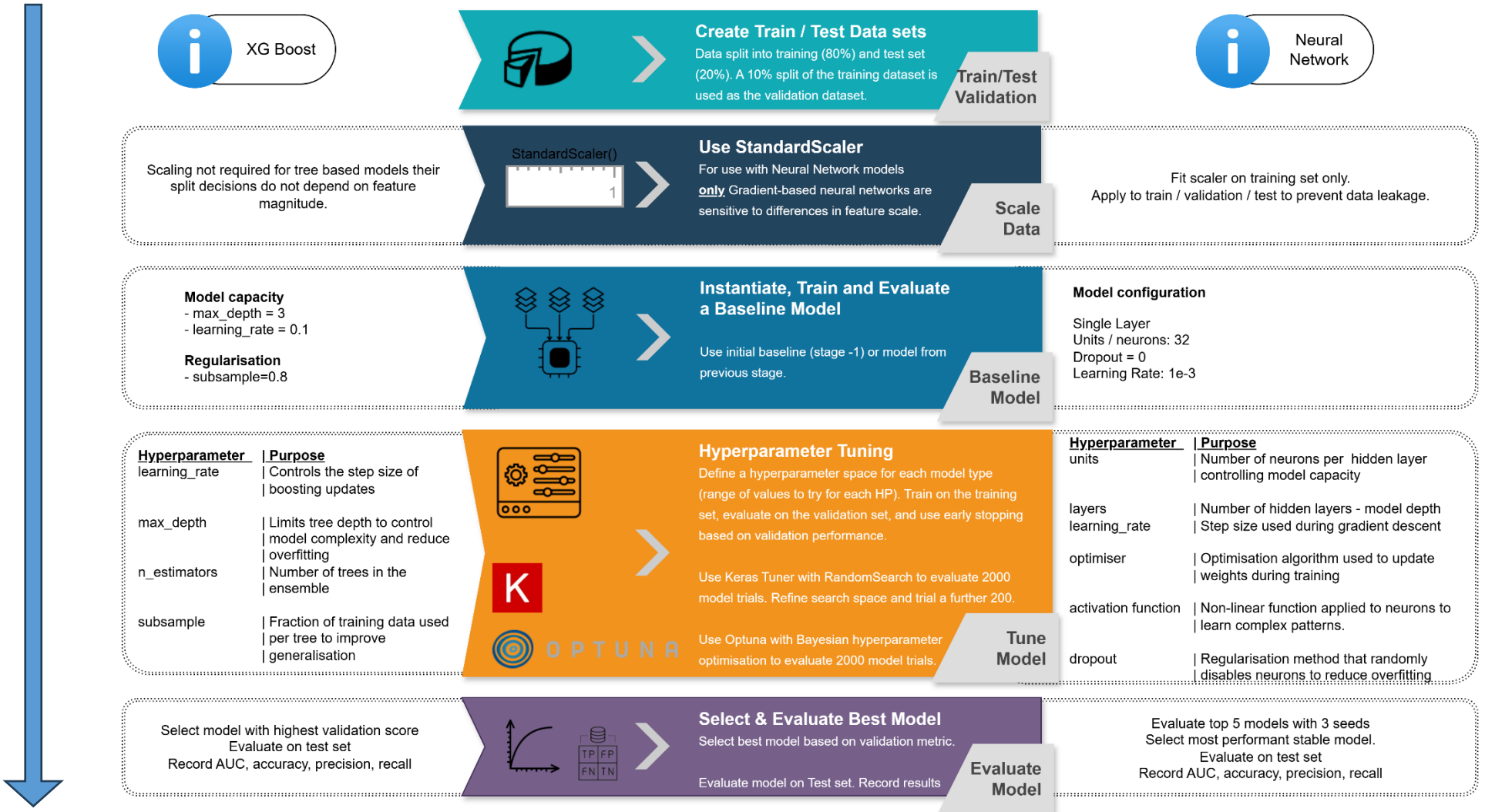


Figure 3 – ML Model Construction and Evaluation

Stage-wise Model Refinement.

Models were retrained at each stage as additional student data became available, allowing comparison of predictive performance across stages.

3.3 Model Evaluation Metrics & Selection Criteria

Model performance was evaluated using the metrics in Table 1. Model stability (neural networks) and simplicity were also considered.

Metric / Criterion	*Purpose*	*Notes*
Accuracy	Overall proportion of correct predictions	Used together with other metrics for model comparison.
Precision	Proportion of predicted dropouts that are correct	Important to avoid false alarms
Recall	Proportion of actual dropouts detected	Important for early intervention
ROC–AUC	Measures classification performance across all decision thresholds	Used for primary model comparison
Confusion Matrix	Shows distribution of TP / FP / TN / FN	Used to interpret prediction errors
Stability (Neural Network)	Measures consistency across multiple runs.	Neural networks may produce different results due to random initialisation
Model Simplicity	Preference for less complex models when performance is similar.	Simpler models are easier to interpret and deploy.

Table 1 - Model Evaluation Metrics

Final performance of the chosen model is evaluated on the held-out test dataset.

Early stopping monitored validation AUC during neural network training and XGBoost tuning, limiting further training when validation performance stopped improving. XGBoost baseline models used 100 estimators without early stopping.

3.4 Feature Importance & Model Interpretation (SHAP)

Feature importance and SHAP were used to interpret model behaviour and compare the effect of additional features across stages.

4 Results

4.1 Baseline Data Analysis

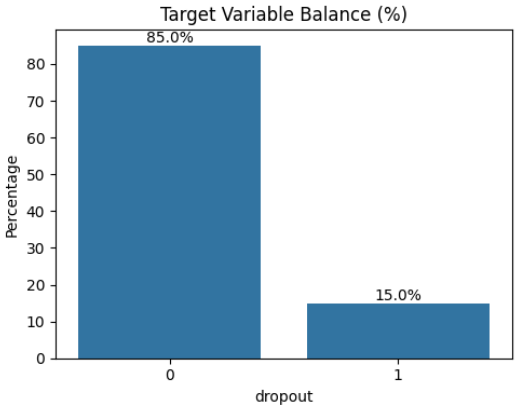


Figure 4 – Target Variable Balance

The target variable is imbalanced (~15% dropout), so ROC–AUC, precision and recall were used in addition to accuracy. Stratified sampling preserved class distribution in train / validation / test splits.

Stage 1

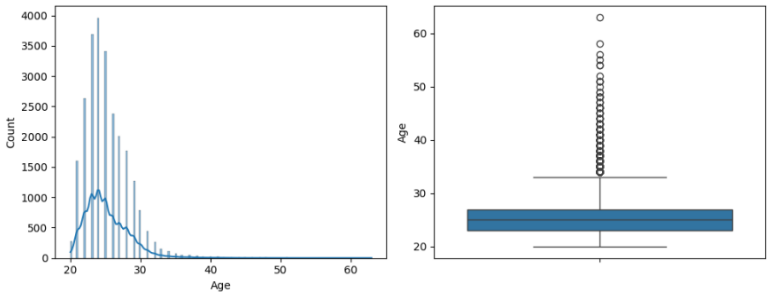


Figure 5 – Stage 1 Data Analysis Visualisation

Age shows moderate right skew but no abnormal values.

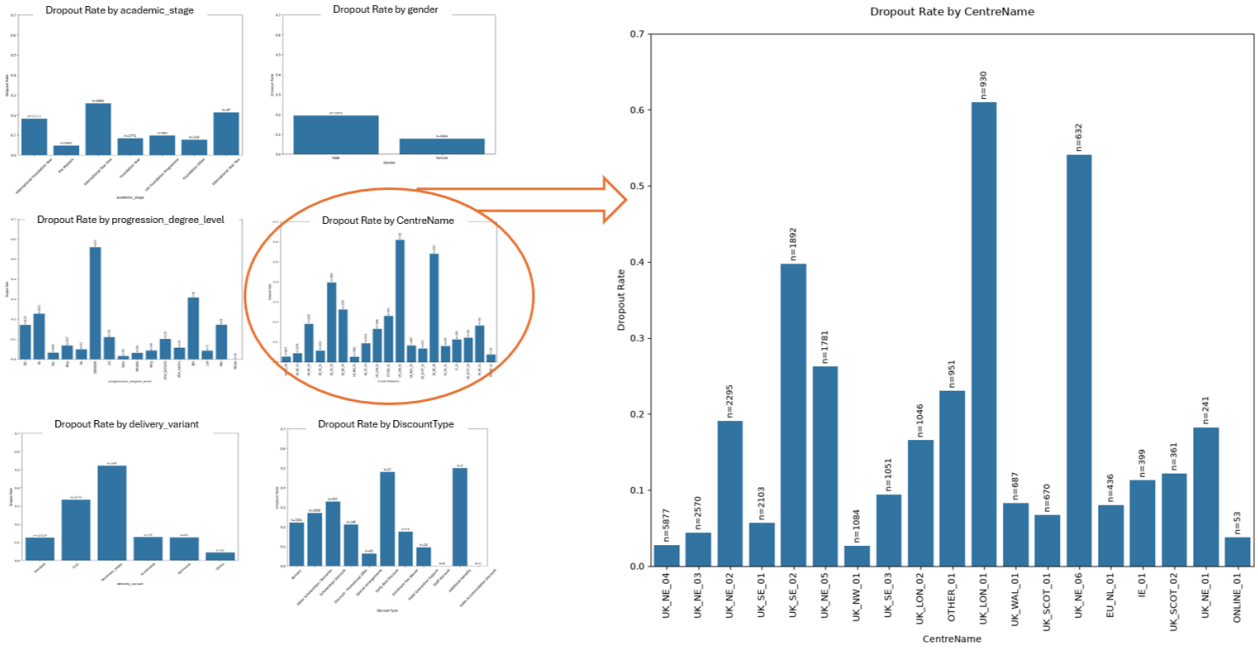


Figure 6 – Stage 1 Data Analysis Visualisation by categorical variables.

Features showing strong separation between categories were retained. For example, CentreName showed large variation in dropout rate and provided strong predictive signal.

Stage 2

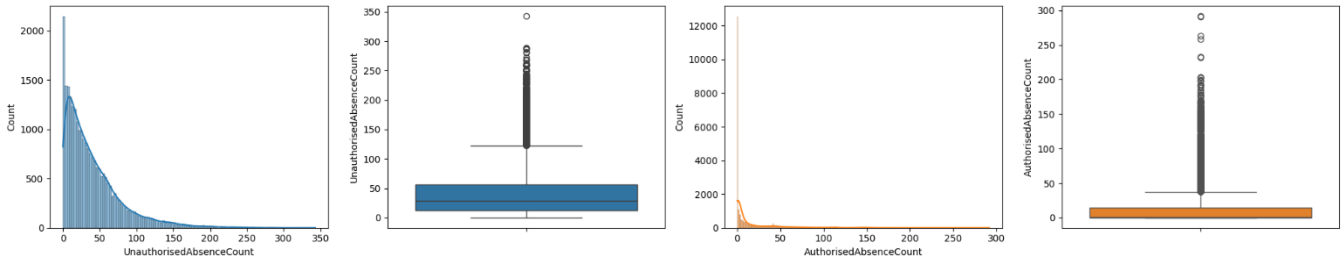
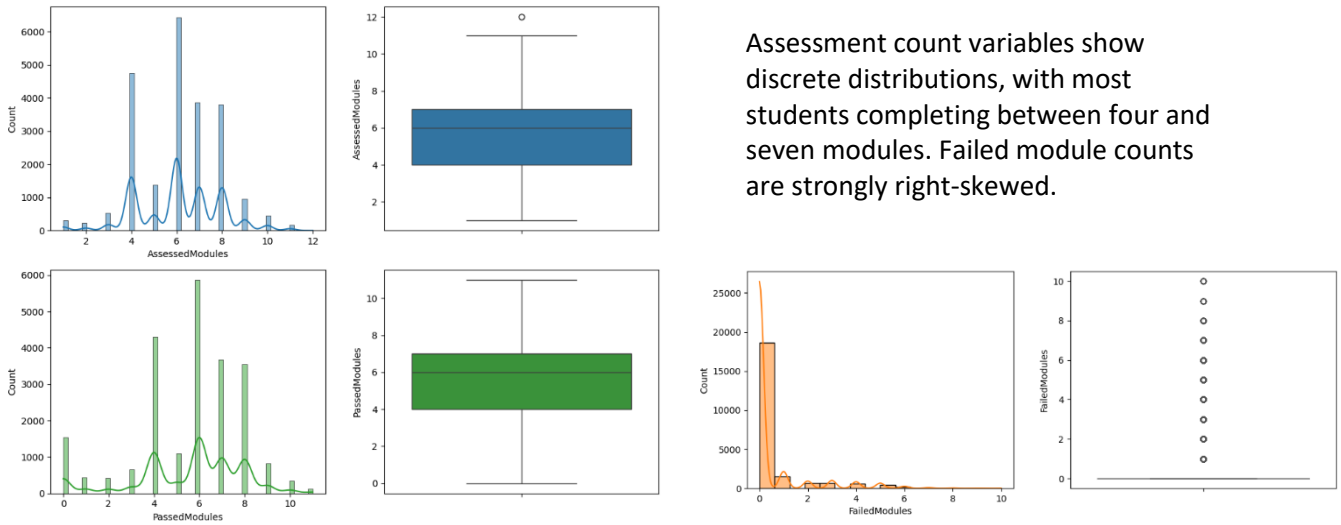


Figure 7 – Stage 2 Data Analysis Visualisation

Absence counts were strongly right-skewed, with higher values associated with increased dropout risk.

Stage 3



Assessment count variables show discrete distributions, with most students completing between four and seven modules. Failed module counts are strongly right-skewed.

Figure 8 – Stage 3 Data Analysis Visualisation

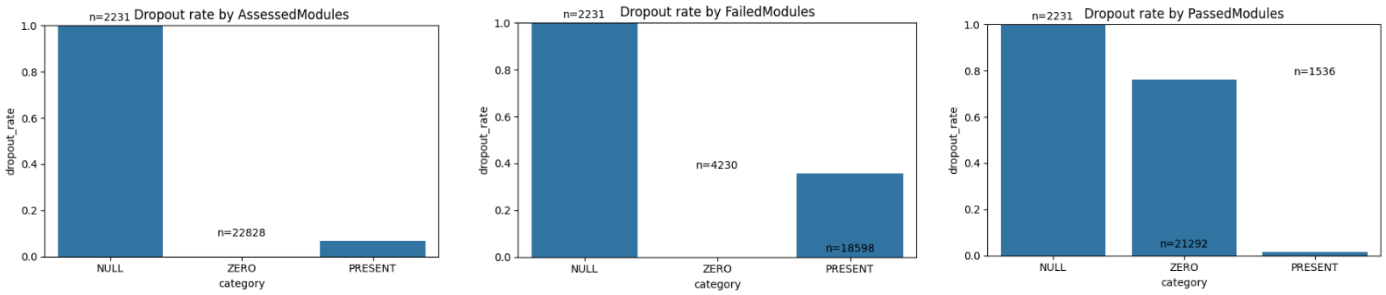


Figure 9 – Stage 3 Data Missingness vs Target

Module count fields contain many missing values, **strongly associated with dropout**.

4.2 Model Performance by Stage

Table 2 shows the performance of the tuned XGBoost and Neural Network models across all dataset stages.

Model	Stage	Accuracy	Precision	Recall	Specificity	ROC–AUC
XGBoost	**Stage 1**	0.8990	0.6979	0.5752	0.9561	0.8988
Neural Network	**Stage 1**	0.8986	0.7063	0.5539	0.9594	0.8978
XGBoost	**Stage 2**	0.9142	0.7627	0.6205	0.9660	0.9326
Neural Network	**Stage 2**	0.9076	0.7345	0.6005	0.9617	0.9182
XGBoost	**Stage 3**	0.9886	0.9819	0.9414	0.9969	0.9989
Neural Network	**Stage 3**	0.9870	0.9738	0.9387	0.9955	0.9986

Table 2 – Performance of tuned XGBoost and Neural Network models across Stages

Performance improves from Stage 1 to Stage 2 and further in Stage 3 for both models, with the largest increase between Stage 2 and Stage 3, indicating strong predictive signal from academic performance features.

Because the target variable is imbalanced, Precision–Recall curves were examined alongside ROC curves. Both confirm improved separation across stages, with Stage 3 achieving near-perfect classification

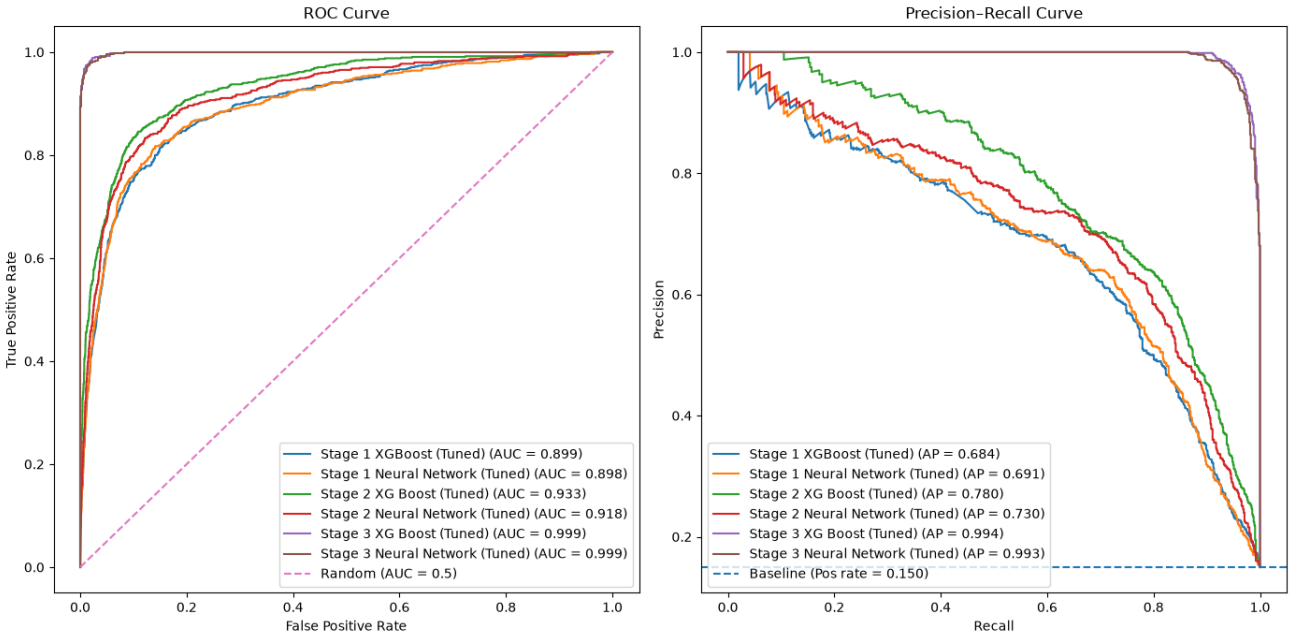


Figure 10 – ROC and PR Curves for Tuned Models by Stage

Performance differences between baseline and tuned models are small compared with the improvement observed between Stage 1, Stage 2, and Stage 3 datasets.

		Neural Network					XG Boost				
Model	Stage	Acc	Prec	Rec	Spec	AUC	Acc	Prec	Rec	Spec	AUC
Baseline	**Stage 1**	0.8951	0.6692	0.5925	0.9484	0.8899	0.8958	0.6951	0.5433	0.9580	0.8951
Tuned	**Stage 1**	▲ 0.8986	▲ 0.7063	▼ 0.5539	▲ 0.9594	▲ 0.8978	▲ 0.8990	▲ 0.6979	▲ 0.5752	▼ 0.9561	▲ 0.8988
Baseline	**Stage 2**	▲ 0.9084	▲ 0.7378	▲ 0.6032	▲ 0.9622	▲ 0.9187	▲ 0.9162	▲ 0.7674	▲ 0.6325	▲ 0.9662	▲ 0.9307
Tuned	**Stage 2**	▼ 0.9076	▼ 0.7345	▼ 0.6005	▼ 0.9617	▼ 0.9182	▼ 0.9142	▼ 0.7627	▼ 0.6205	▼ 0.9660	▲ 0.9326
Baseline	**Stage 3**	▲ 0.9850	▲ 0.9815	▲ 0.9174	▲ 0.9969	▲ 0.9981	▲ 0.9884	▲ 0.9766	▲ 0.9454	▲ 0.9960	▲ 0.9989
Tuned	**Stage 3**	▲ 0.9870	▼ 0.9738	▲ 0.9387	▼ 0.9955	▲ 0.9986	▲ 0.9886	▲ 0.9819	▼ 0.9414	▲ 0.9969	0.9989

Table 3 – Effect of model Hyperparameter Tuning (Baseline vs Tuned) at each stage.

Across all stages, hyperparameter tuning results in only small performance changes, while the introduction of additional features with stronger predictive signal leads to much larger improvements, indicating that feature quality is the primary driver of model accuracy in this dataset.

4.3 Effect of Additional Features (Stage 2 & 3)

SHAP plots for the top 10 features were used to identify the variables with the greatest impact at each stage.

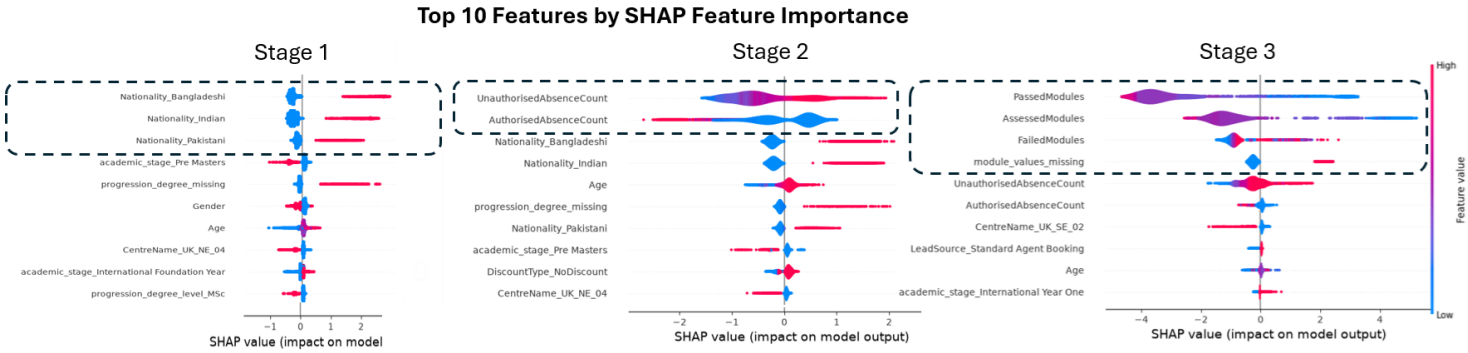


Figure 11 – Top 10 features by SHAP Feature Importance at each stage.

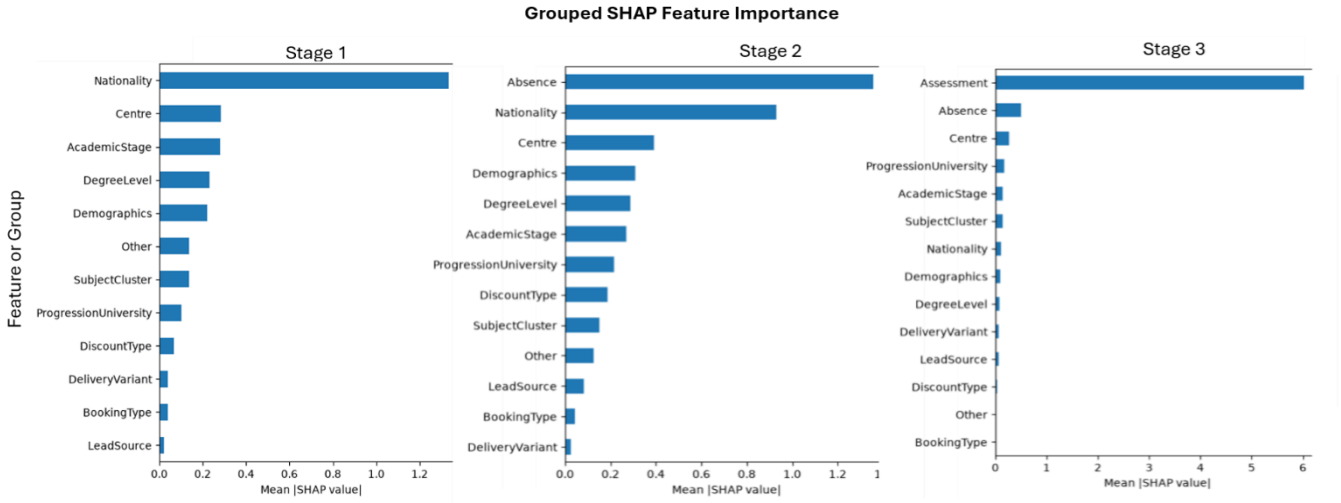


Figure 12 – Mean |SHAP value| for all features grouped by category

Stage	*Most Important Features*	*Interpretation*	*Predictive Reliability*
1	Nationality, CentreName, DegreeLevel, AcademicStage	Prediction based mainly on student background and course characteristics. Limited predictive power at early stage.	◔ Low
2	Absence counts, Nationality, CentreName	Attendance behaviour provides stronger signal, improving identification of disengaged students.	◑ Medium
3	AssessedModules, PassedModules, FailedModules	Academic performance dominates prediction, explaining large increase in model accuracy.	● Very High

Table 4 – Feature Importance Interpretation per Stage

Outcomes are consistently influenced by Centre across all stages, indicating that institutional differences contribute to variation in student progression. The feature importance analysis shows that centre-related features remain predictive even after strongly predictive additional behavioural and academic variables are introduced.

Predictive performance increases across stages as attendance and academic information is added. Whether this improvement supports earlier intervention depends on when those features become available relative to withdrawal.

4.4 Final Model Comparison

Model performance was compared across the three dataset stages to evaluate the effect of additional information on predictive accuracy.

Aspect	*XGBoost*	*Neural Network*
Predictive performance	Higher test ROC AUC at each stage	Similar performance to XGBoost in Stage 3
Recall at threshold 0.5	Higher recall in Stages 1–3 for the selected final models	Slightly lower recall at each stage
Effect of additional features	Clear improvement with attendance and assessment data	Clear improvement with attendance and assessment data
Effect of further tuning	Small changes, with trade-offs between metrics	Small changes; final Stage 2 configuration did not improve on its baseline
Preprocessing	Used unscaled numeric and encoded categorical features	Required standardised numeric and encoded categorical inputs
Interpretation	Gain-based feature importance and SHAP used to explain predictions	Equivalent model-level interpretation was not included in this analysis
Practical use	Candidate for further operational evaluation	Alternative candidate, particularly with Stage 3 data

Table 5 – Comparison of XGBoost and Neural Network models

XGBoost achieved higher test AUC than the selected neural network at each stage, although their performance was very similar in Stage 3.

In this context, recall is particularly important because missed dropout cases represent missed opportunities for support. Its priority relative to precision depends on the costs of missed cases, unnecessary interventions and available support capacity.

5 Conclusions & Next Steps

Both model types identify dropout cases in the supplied dataset, with performance improving as attendance and academic information is added. XGBoost achieved higher test AUC at each stage and is a strong candidate for further operational evaluation.

The staged approach could support progressively updated risk estimates, from enrolment through attendance monitoring to assessment results. Practical usefulness depends on confirming feature availability at each prediction point and evaluating whether predictions provide sufficient advance warning for intervention.

Recommended next steps:

- Evaluate early-stage models for student screening and induction planning, using only information confirmed as available at enrolment.
- Capture attendance data over time and assess whether mid-course predictions identify disengagement early enough for useful intervention.
- Evaluate the additional benefit of assessment information as results become available.
- Compare Stage 3 predictions with simple rules based directly on assessment results. If these results already reveal non-completion, direct monitoring may be more practical than predictive modelling.
- Investigate the differences associated with study centre, including student mix, course provision and internal processes. Feature importance identifies predictive associations, but does not establish their causes or show that standardising processes would improve progression.

Appendices

Appendix A – Stage 1 Feature Summary

A.1 Feature Review Table (full audit)

FEATURE	SOURCE	TYPE	ENGINEERED	USED	DECISION	NOTES
LEARNERCODE	Original	ID	No	No	Drop	Identifier only
COMPLETEDCOURSE	Original	Target	No	Yes	Target	Converted to binary target
DATEOFBIRTH	Original	Date	No	No	Transform	Used to derive Age
AGE	Engineered	Numeric	Yes	Yes	Retain	Derived from DateOfBirth
GENDER	Original	Binary	No	Yes	Retain	Encoded 0/1
NATIONALITY	Original	Categorical	No	No	Transform	Grouped to reduce cardinality
NATIONALITY_GROUPED	Engineered	Categorical	Yes	Yes	Retain	Rare categories grouped
COURSENAME	Original	Categorical	No	No	Transform	Used to derive subject_cluster
SUBJECT_CLUSTER	Engineered	Categorical	Yes	Yes	Retain	Reduced cardinality
COURSELEVEL	Original	Categorical	No	No	Drop	Captured by academic_stage
ACADEMIC_STAGE	Engineered	Categorical	Yes	Yes	Retain	Derived from CourseLevel
DELIVERY_VARIANT	Engineered	Categorical	Yes	Yes	Retain	Derived from intake info
CENTRENAME	Original	Categorical	No	Yes	Retain	Moderate cardinality
BOOKINGTYPE	Original	Categorical	No	Yes	Retain	Low cardinality
LEADSOURCE	Original	Categorical	No	Yes	Retain	Moderate signal
DISCOUNTTYPE	Original	Categorical	No	Yes	Retain	Null → NoDiscount
HOMECITY	Original	Categorical	No	No	Drop	High cardinality, missing
HOMESTATE	Original	Categorical	No	No	Drop	High cardinality, missing
PROGRESSIONDEGREE	Original	Categorical	No	No	Transform	Used to derive degree level
PROGRESSION_DEGREE_LEVEL	Engineered	Categorical	Yes	Yes	Retain	Collapsed categories
PROGRESSION_DEGREE_MISSING	Engineered	Binary	Yes	Yes	Retain	Missing indicator for ProgressionDegree
PROGRESSIONUNIVERSITY	Original	Categorical	No	Yes	Retain	Rare categories grouped
ISFIRSTINTAKE	Original	Binary	No	Yes	Retain	Behavioural feature

Table A.1 – Comparison of XGBoost and Neural Network models

A.2 Imputation Rules

Missing values were reviewed during preprocessing.

Imputation was applied only where missingness represented a meaningful category rather than data loss.

FEATURE	MISSING %	RULE APPLIED	NOTES
DISCOUNTTYPE	69.69%	Replace null with "NoDiscount"	Missing interpreted as absence of discount

A.3 Final Feature List (Pre-Encoding)

The following features were used as model inputs after preprocessing and feature engineering, and before one-hot encoding was applied.

FEATURE	TYPE	ENGINEERED	NOTES
CENTRENAME	Categorical	No	
BOOKINGTYPE	Categorical	No	
LEADSOURCE	Categorical	No	
DISCOUNTTYPE	Categorical	No	Missing values present
GENDER	Categorical	No	Binary after encoding
NATIONALITY	Categorical	No	High cardinality
ISFIRSTINTAKE	Boolean	No	Converted to int
PROGRESSIONUNIVERSITY	Categorical	No	
AGE	Numeric	Yes	Derived from DateOfBirth
PROGRESSION_DEGREE_LEVEL	Categorical	Yes	Derived from ProgressionDegree
ACADEMIC_STAGE	Categorical	Yes	Derived from CourseLevel
SUBJECT_CLUSTER	Categorical	Yes	Derived from CourseName
DELIVERY_VARIANT	Categorical	Yes	Derived from CourseName
PROGRESSION_DEGREE_MISSING	Numeric	Yes	Missing indicator
DROPOUT	Target	Yes	Binary target variable

A.4 Encoding Strategy

Categorical features were encoded prior to modelling.

One-hot encoding was used for multi-category variables, and binary encoding was used for two-class variables.

FEATURE	TYPE	ENCODING	NOTES
ACADEMIC_STAGE	Categorical	One-hot	—
SUBJECT_CLUSTER	Categorical	One-hot	—
DELIVERY_VARIANT	Categorical	One-hot	—
CENTRENAME	Categorical	One-hot	—
NATIONALITY (GROUPED)	Categorical	One-hot	Rare categories grouped before encoding
PROGRESSION_DEGREE_LEVEL	Categorical	One-hot	—
PROGRESSIONUNIVERSITY	Categorical	One-hot	Rare categories collapsed
DISCOUNTTYPE	Categorical	One-hot	Null → NoDiscount before encoding
BOOKINGTYPE	Categorical	One-hot	—
LEADSOURCE	Categorical	One-hot	—
GENDER	Binary	Binary (0/1)	Male=0, Female=1
ISFIRSTINTAKE	Binary	Binary (0/1)	False=0, True=1
AGE	Numeric	None	Used as numeric feature

Feature Counts

- Total features before encoding: 14
- Total features after encoding: 174

Appendix B – Stage 2 Feature Summary (additional features)

B.1 Feature Review Table

FEATURE	SOURCE	TYPE	ENGINEERED	USED	DECISION	NOTES
AUTHORISEDABSENCECOUNT	Stage 2	Numeric	No	Yes	Retain	Count of authorised absences across modules
UNAUTHORISEDABSENCECOUNT	Stage 2	Numeric	No	Yes	Retain	Count of unauthorised absences across modules
ABSENCE_VALUES_MISSING	Engineered	Binary	Yes	Yes	Retain	Value = 1 when absence features are missing

B.2 Imputation Rules

FEATURE	TYPE	HAS MISSING	IMPUTATION METHOD	REASON
AUTHORISEDABSENCECOUNT	Numeric	Yes	Fill with 0	Zero placeholder; missingness retained by absence_values_missing.
UNAUTHORISEDABSENCECOUNT	Numeric	Yes	Fill with 0	Zero placeholder; missingness retained by absence_values_missing.
ABSENCE_VALUES_MISSING	Binary	No	Derived	Indicates original absence values were missing

B.2.1 Missing Indicator Justification

Missing Indicator Encoding Logic:

ABSENCE_VALUES_MISSING	AUTHORISEDABSENCECOUNT	MEANING
0	5	Real value
0	0	Real zero
1	0	Missing → imputed

Exploratory analysis of the absence variables showed that records with missing values had a significantly higher dropout rate than those with zero or non-zero absence counts. This indicates that missing values do not represent true zeros, but instead correspond to a different underlying data condition.

For this reason, absence counts were imputed with zero while also introducing a binary indicator feature (absence_values_missing) to preserve the information that the original value was missing. This allows the model to distinguish between true zero absences and imputed values.

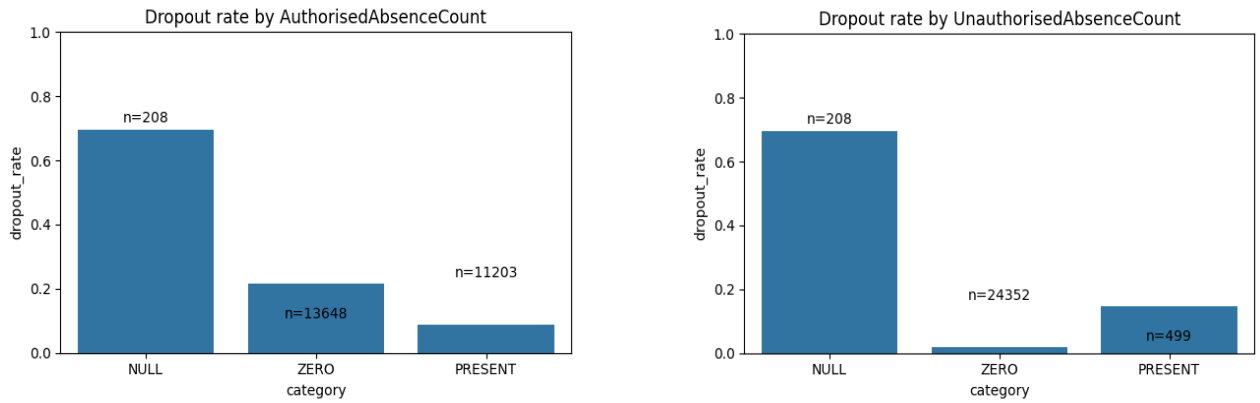


Figure B.1 – Dropout rate by absence count category (missing / zero / present)

Appendix C – Stage 3 Feature Summary (additional features)

C.1 Feature Review Table

FEATURE	SOURCE	TYPE	ENGINEERED	USED	DECISION	NOTES
ASSESSEDMODULES	Stage 3 dataset	Numeric	No	Yes	Retain	Count of assessed modules
PASSEDMODULES	Stage 3 dataset	Numeric	No	Yes	Retain	Count of passed modules
FAILEDMODULES	Stage 3 dataset	Numeric	No	Yes	Retain	Count of failed modules
MODULE_VALUES_MISSING	Engineered	Binary	Yes	Yes	Retain	Indicator for missing module count values

C.2 Imputation Rules

FEATURE	TYPE	HAS MISSING	IMPUTATION METHOD	REASON
ASSESSEDMODULES	Numeric	Yes	Fill with 0	Missing interpreted as no recorded module values
PASSEDMODULES	Numeric	Yes	Fill with 0	Missing interpreted as no recorded module values
FAILEDMODULES	Numeric	Yes	Fill with 0	Missing interpreted as no recorded module values
MODULE_VALUES_MISSING	Binary	No	Derived	Indicates original module values were missing

C.2.1 Missing Indicator Justification

Missing Indicator Encoding Logic

MODULE_VALUES_MISSING	FAILEDMODULES	INTERPRETATION
0	2	Real value
0	0	Real zero
1	0	Missing → imputed

Exploratory analysis of the module count variables showed that records with missing values had significantly higher dropout rates than records with valid module counts. This indicates that missing values do not represent true zeros, but instead correspond to a different underlying data condition.

Removing these rows would discard a large proportion of the dataset and remove informative signal. For this reason, module count values were imputed with zero while also introducing a binary indicator feature (module_values_missing) to preserve the information that the original values were missing.

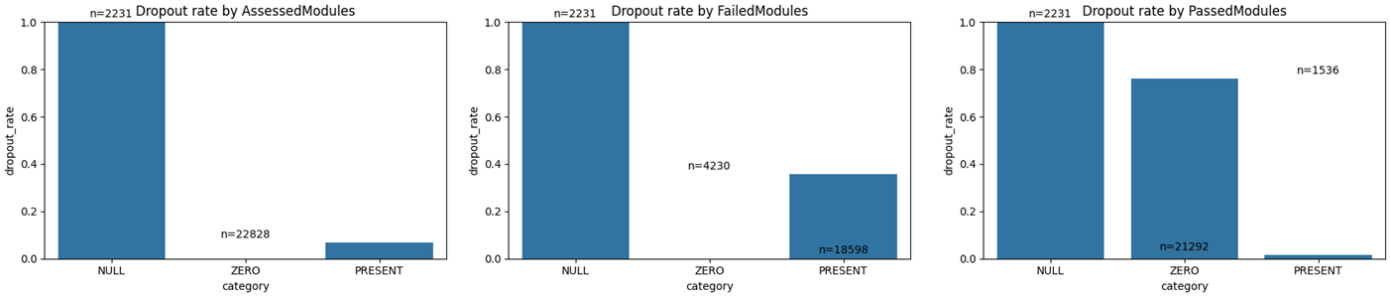


Figure C.1 – Dropout rate by module result count category (missing / zero / present)

Appendix D – Hyperparameters Tuning

D.1 XGBoost Search Space

HYPERPARAMETER	SEARCH RANGE	BASELINE	JUSTIFICATION
LEARNING_RATE	0.01 – 0.20	0.10	Covers slower boosting through to baseline-fast learning
MAX_DEPTH	3 – 8	3	Explores shallow to moderate tree depth
MIN_CHILD_WEIGHT	1 – 12	default	Controls minimum leaf weight, higher values regularise
GAMMA	0 – 5	default	Penalises splits, encourages simpler trees
SUBSAMPLE	0.60 – 1.00	0.80	Row sampling regularisation
COLSAMPLE_BYTREE	0.60 – 1.00	0.80	Feature sampling regularisation
REG_ALPHA	1e-8 – 10	default≈0	L1 regularisation for wide tabular data
REG_LAMBDA	1e-2 – 50	default≈1	L2 regularisation strength
N_ESTIMATORS	3000 (fixed)	100	Set high, early stopping used
EARLY_STOPPING_ROUNDS	30 (fixed)	–	Prevents overfitting, replaces tuning n_estimators

The XGBoost search space was not refined after initial tuning.

The selected ranges provided stable performance across all stages, and early stopping was used to determine the effective number of trees.

D.2 Neural Network Search Space

HYPERPARAMETER	SEARCH RANGE / OPTIONS	NOTES
UNITS	16 – 256 (step 16)	Neurons per hidden layer
LAYERS	1 – 4	Number of hidden layers
DROPOUT	0.0 – 0.5 (step 0.1)	Dropout after hidden layers
ACTIVATION	relu, tanh, elu	Hidden layer activation
OPTIMIZER	adam, rmsprop	Optimisation algorithm
LEARNING_RATE	1e-2, 3e-3, 1e-3, 3e-4	Learning rate
L2_STRENGTH	0.0, 1e-6, 1e-5, 1e-4, 1e-3	L2 regularisation strength
BATCH_SIZE	64	Training batch size
EARLY_STOPPING	patience=5	Monitor val_auc, restore_best_weights=True, min_delta=1e-4

Training used early stopping on validation AUC with patience = 5, and the best model weights were restored after training. Batch size was fixed to 64 for all experiments.

D.3 Neural Network Validation AUC Distribution by Hyperparameter

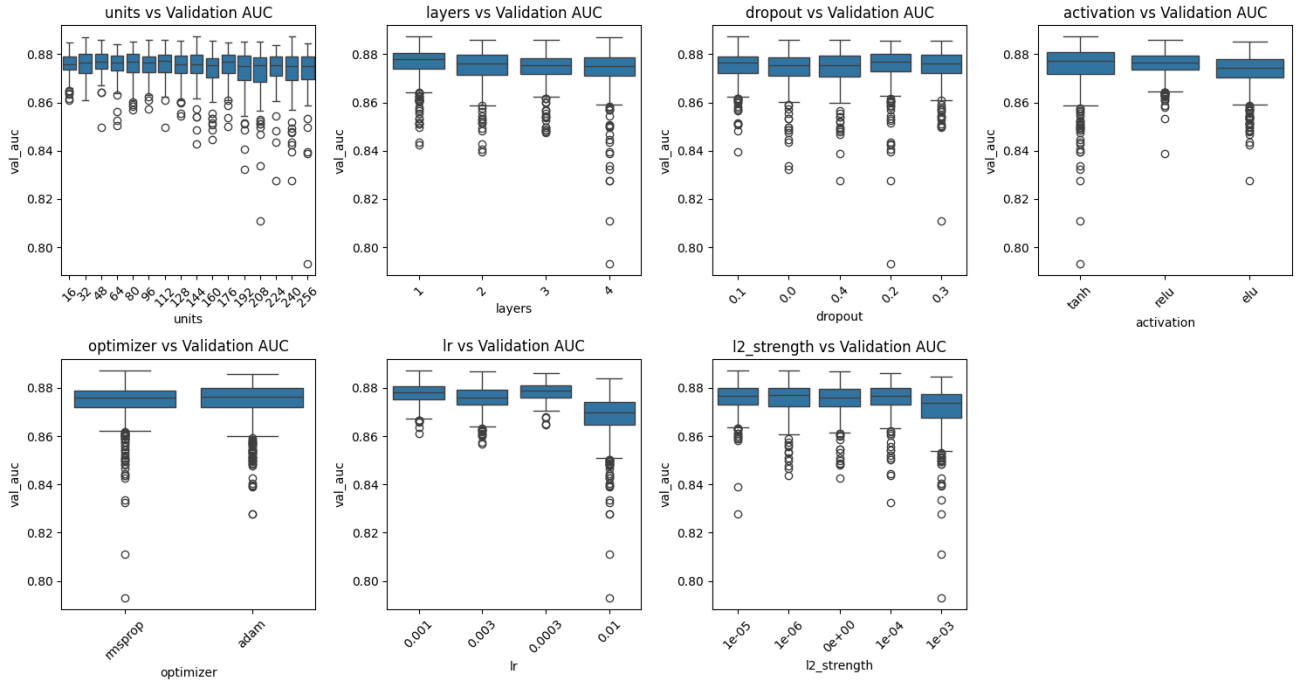


Figure D.1 – Validation AUC Distribution by hyperparameter

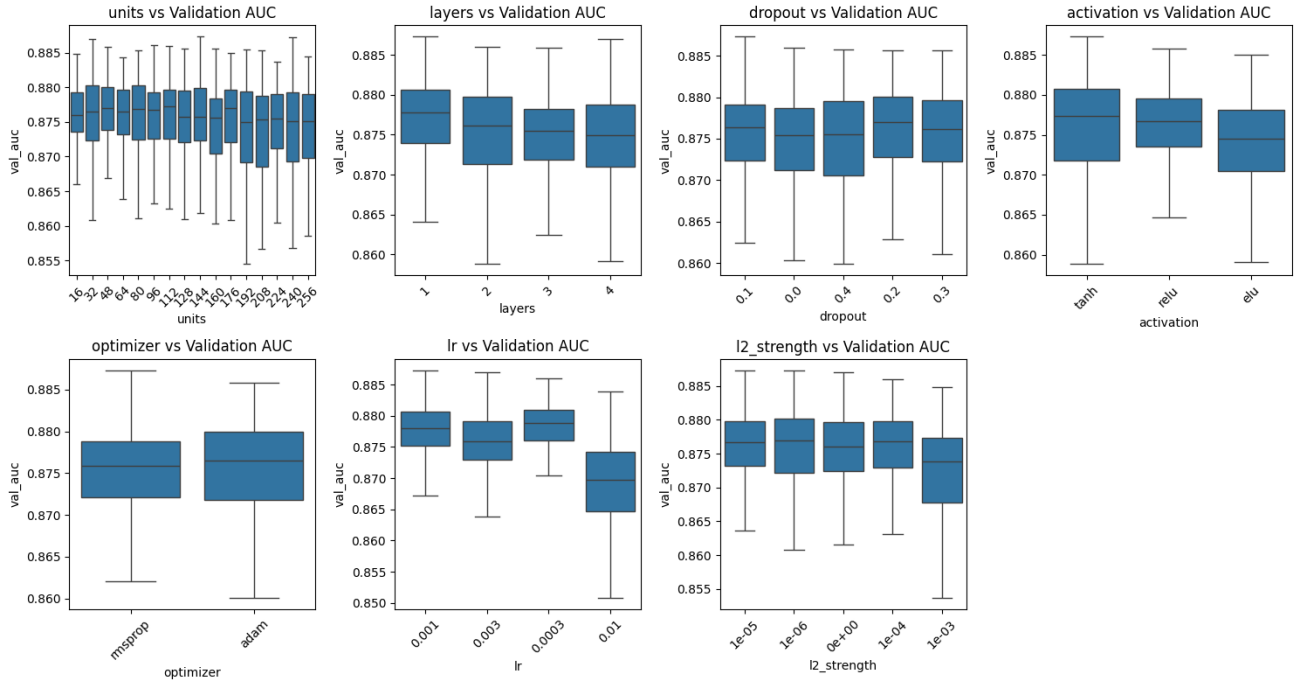


Figure D.2 – Validation AUC Distribution by hyperparameter – outliers not shown

Validation AUC scores were recorded for each neural network hyperparameter configuration during tuning. The box plots below show the distribution of validation AUC across configurations for each hyperparameter. The first set of plots shows all trials.

The second set shows the same data with extreme outliers removed to improve visual clarity and allow comparison of the main distribution.

D.4 Surrogate Model SHAP Plots

Additional insight into the tuning process was obtained using SHAP analysis on a surrogate regression tree model, with validation AUC as the target variable and the hyperparameters as input features.

D.4.1 XG Boost

Although no refinement of the hyperparameter search space was required for the XGBoost models, SHAP analysis of the surrogate tuning model shows clear structure in the parameter importance.

This behaviour is consistent with the Bayesian optimisation strategy used by Optuna, where the search progressively focuses on promising regions of the parameter space rather than sampling uniformly.

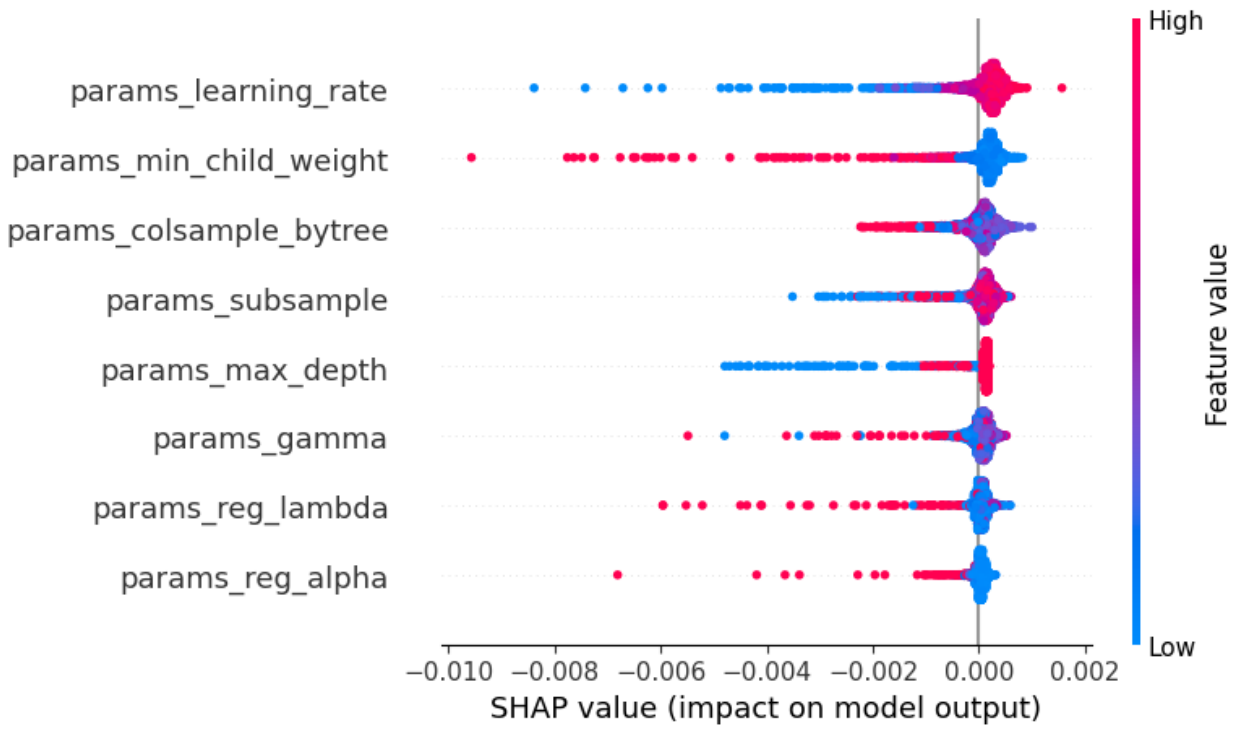


Figure D.3 – SHAP analysis of surrogate model trained against XGBoost hyperparameters

D.4.1 Neural Network

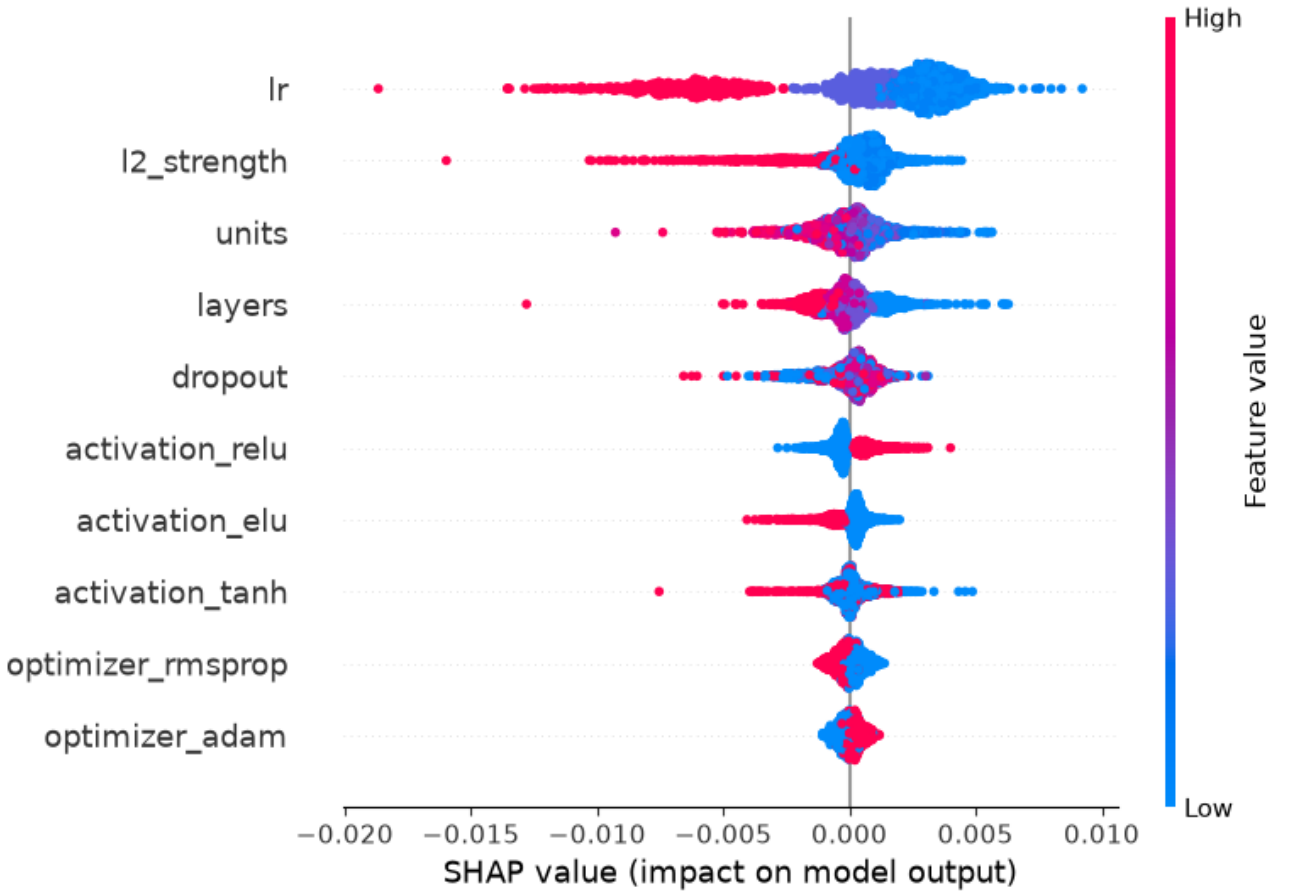


Figure D.4 – Hyperparameter importance from surrogate model (neural network tuning)

The SHAP summary plot shows that learning rate, L2 regularisation strength, and network size parameters (units and layers) have the largest impact on validation AUC, while activation function and optimizer choice contribute relatively little to performance variation.

D.5 Neural Network Refined Search Space

Based on the validation AUC distributions, SHAP analysis, and tabulated tuning results, the neural network search space was reduced to focus on stable regions of the parameter space.

HYPERPARAMETER	INITIAL RANGE	REFINED RANGE	EVIDENCE SUMMARY
UNITS	16–256	32–192 (step 16)	Similar performance across small–medium models; very large models show higher variance
LAYERS	1–4	1–4	Performance similar across depths; no consistent benefit from deeper networks
DROPOUT	0.0–0.5	0.0–0.4 (step 0.1)	Moderate dropout performs best; extreme values not beneficial
ACTIVATION	relu, tanh, elu	relu, tanh	ELU shows lower median validation AUC
OPTIMIZER	adam, rmsprop	adam, rmsprop	Both optimizers stable; no clear difference
LEARNING_RATE	1e-2, 3e-3, 1e-3, 3e-4	3e-4, 1e-3	Higher learning rates reduce stability
L2_STRENGTH	0.0–1e-3	0.0, 1e-6, 1e-5, 1e-4	Strong regularisation reduces performance

Total combinations in refined search space: 11 × 4 × 5 × 2 × 2 × 2 × 4 = 7040

D.6 Tuning Procedure

Hyperparameter tuning was performed using Optuna for XGBoost and Keras Tuner RandomSearch for neural networks.

For each trial, models were trained on the training set and evaluated on the validation set using validation AUC as the optimisation metric.
The validation set was not used for final performance reporting.

Neural network

For the neural network, early stopping was applied during training with the following settings:

- monitor = validation AUC
- patience = 5
- min_delta = 1e-4
- restore_best_weights = True

Batch size was fixed to 64 for all neural network experiments.

The initial neural network search space was intentionally broad to explore a wide range of architectures.

After reviewing validation AUC distributions, SHAP analysis of the surrogate tuning model, and tabulated results, the neural network search space was reduced to focus on stable regions of the parameter space.

The refined search space was then used for tuning in later stages.

XGBoost

For XGBoost, the number of estimators was set to a high value (3000), and early stopping with 30 rounds was used to determine the effective number of trees during training.

Stage-wise re-tuning

For each stage, the final model was retrained using the best hyperparameters found during tuning before evaluation on the test set.

Appendix E – Final Model Hyperparameter Values

The final hyperparameter configurations for each stage and model are shown below.

E.1 XG Boost

STAGE	MODEL	TYPE	LEARNING _RATE	MAX _DEPTH	SUBSAMPLE	COLSAMPLE _BYTREE	MIN_CHILD _WEIGHT	GAMMA	REG_ALPHA	REG_LAMBDA
STAGE 1	XGBoost	Baseline	0.100000	3	0.800000	0.800000	1	0	0	1
STAGE 1	XGBoost	Tuned	0.146668	8	0.856013	0.988280	1.107940	1.868997	0.019465	0.030819
STAGE 2	XGBoost	Baseline	0.146668	8	0.856013	0.988280	1.107940	1.868997	0.019465	0.030819
STAGE 2	XGBoost	Tuned	0.119	8	0.754	0.601	1.15	0.27	0.0052	4.74
STAGE 3	XGBoost	Baseline	0.119	8	0.754	0.601	1.15	0.27	0.0052	4.74
STAGE 3	XGBoost	Tuned	0.200	8	0.932	0.612	1.18	0.67	0.00000002	0.064

E.2 Neural Network

STAGE	MODEL	TYPE	UNITS	LAYERS	DROPOUT	LEARNING_RATE	L2_STRENGTH	OPTIMIZER	ACTIVATION
STAGE 1	Neural network	Baseline	32	1	0.0	0.001	0	adam	relu
STAGE 1	Neural network	Tuned	240	3	0.3	0.0003	0.0001	adam	relu
STAGE 2	Neural network	Baseline	240	3	0.3	0.0003	0.0001	adam	relu
STAGE 2	Neural network	Tuned	160	3	0.3	0.001	0	adam	relu
STAGE 3	Neural network	Baseline	160	3	0.3	0.001	0	adam	relu
STAGE 3	Neural network	Tuned	160	2	0.1	0.001	0.00001	adam	tanh

Appendix F – Detailed Model Performance Metrics

STAGE	MODEL	TYPE	ACCURACY	PRECISION	RECALL	SPECIFICITY	AUC
STAGE 1	XGBoost	Baseline	0.8958	0.6951	0.5433	0.9580	0.8951
STAGE 1	XGBoost	Tuned	0.8990	0.6979	0.5752	0.9561	0.8988
STAGE 1	Neural network	Baseline	0.8951	0.6692	0.5925	0.9484	0.8899
STAGE 1	Neural network	Tuned	0.8986	0.7063	0.5539	0.9594	0.8978
STAGE 2	XGBoost	Baseline	0.9162	0.7674	0.6325	0.9662	0.9307
STAGE 2	XGBoost	Tuned	0.9142	0.7627	0.6205	0.9660	0.9326
STAGE 2	Neural network	Baseline	0.9084	0.7378	0.6032	0.9622	0.9187
STAGE 2	Neural network	Tuned	0.9076	0.7345	0.6005	0.9617	0.9182
STAGE 3	XGBoost	Baseline	0.9884	0.9766	0.9454	0.9960	0.9989
STAGE 3	XGBoost	Tuned	0.9886	0.9819	0.9414	0.9969	0.9989
STAGE 3	Neural network	Baseline	0.9850	0.9815	0.9174	0.9969	0.9981
STAGE 3	Neural network	Tuned	0.9870	0.9738	0.9387	0.9955	0.9986

Appendix G – Additional Plots

G.1 Stage 1 XGBoost Performance Curves

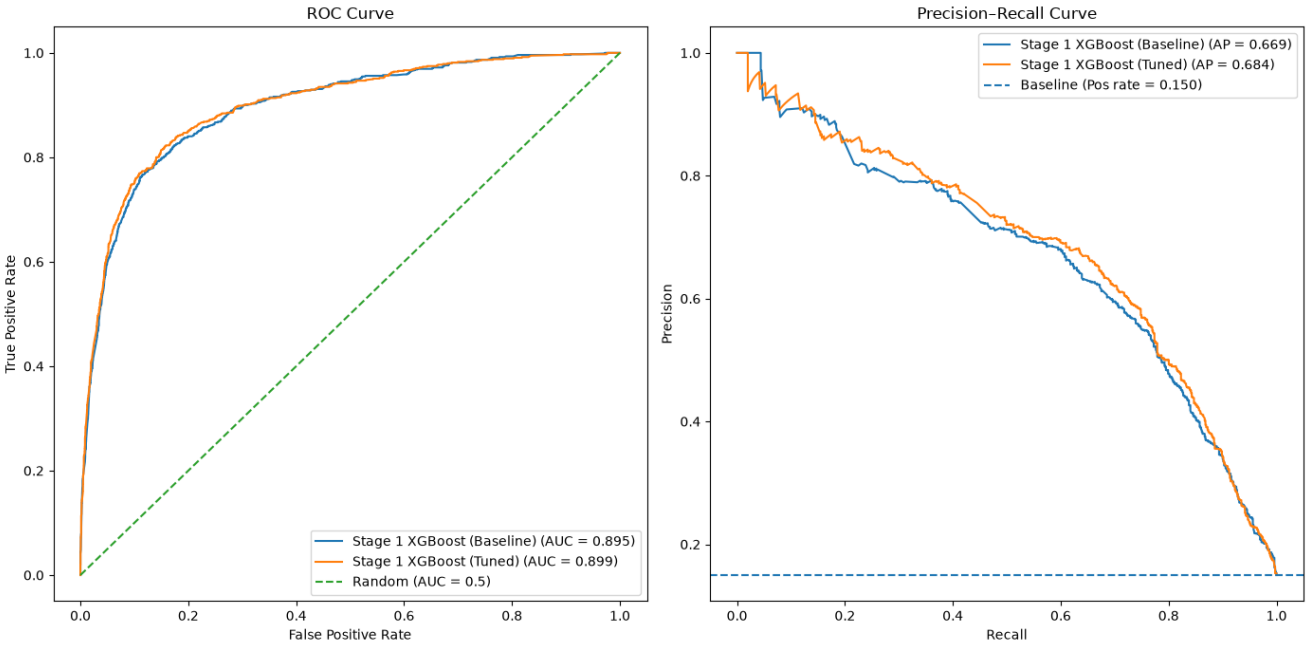


Figure G.1 – XGBoost Performance Curves – Stage 1

G.2 Stage 2 XGBoost Performance Curves

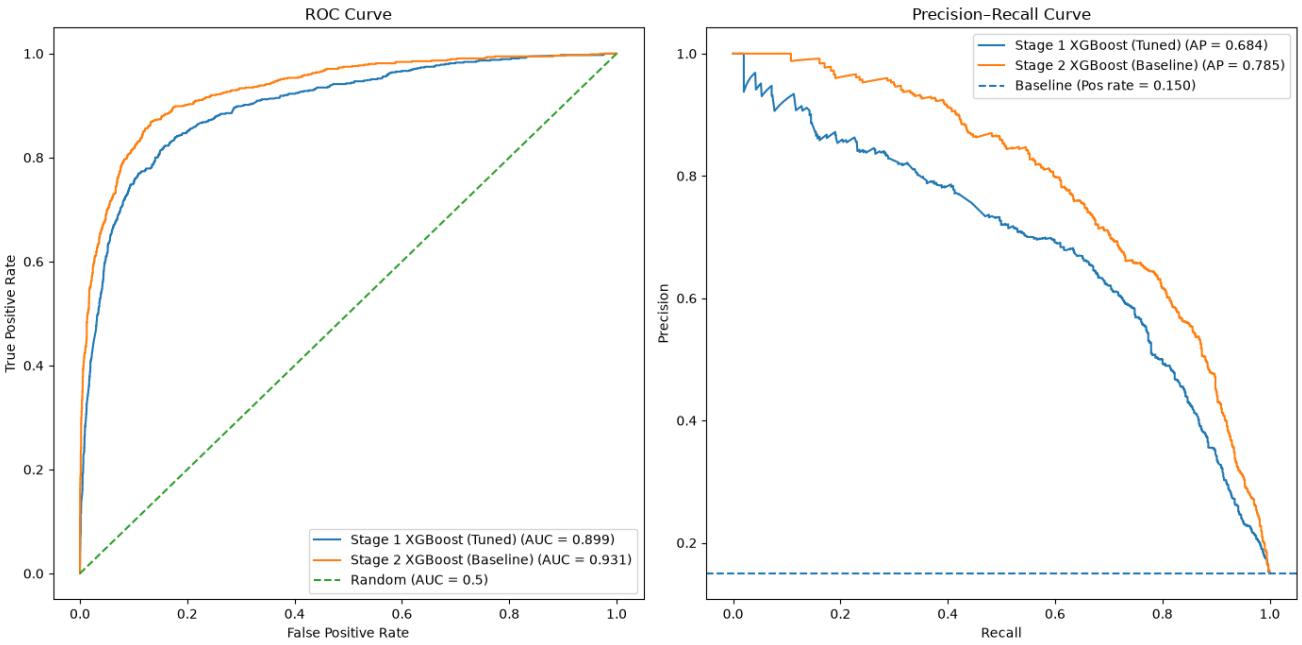


Figure G.2 – XGBoost Performance Curves – Stage 2 Baseline vs Stage 1

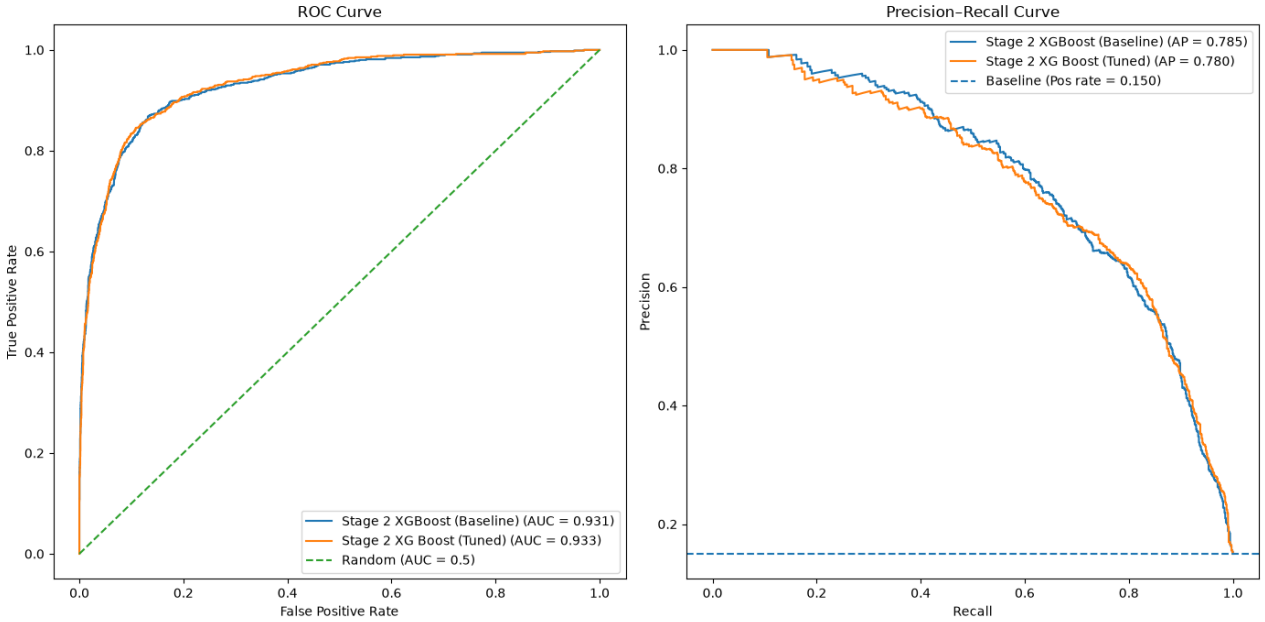


Figure G.3 – XGBoost Performance Curves – Stage 2

G.3 Stage 3 XGBoost Performance Curves

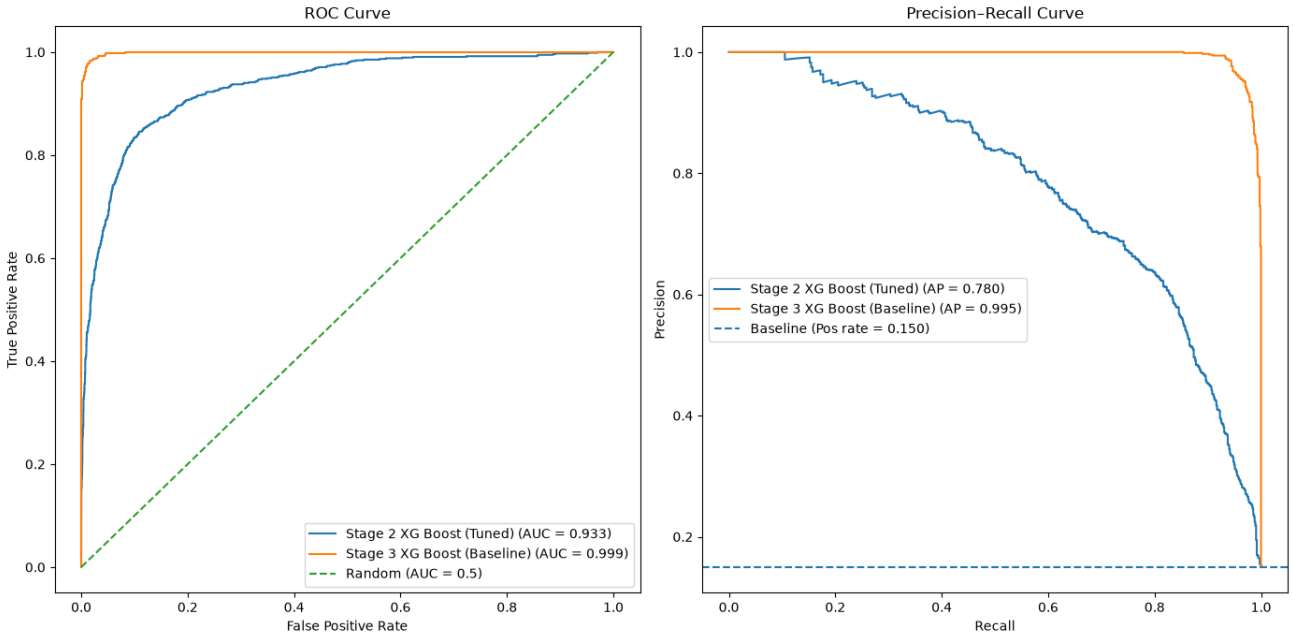


Figure G.4 – XGBoost Performance Curves – Stage 3 Baseline vs Stage 2

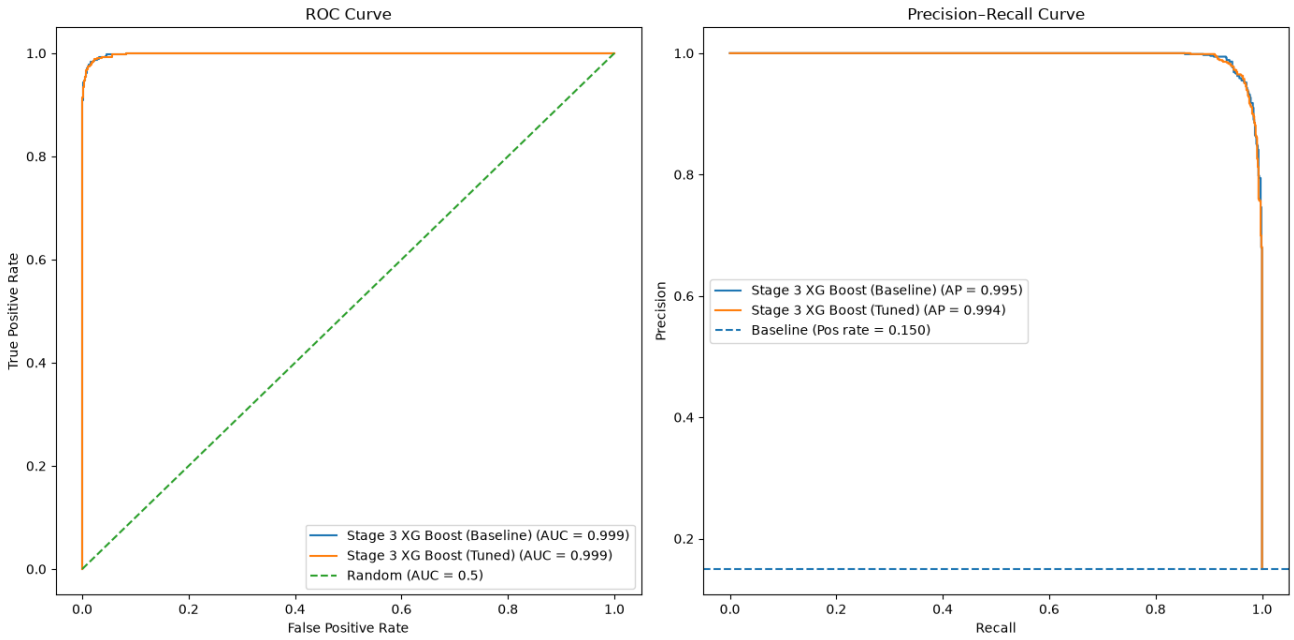


Figure G.4 – XGBoost Performance Curves – Stage 3

G.5 Stage 1 Neural Network Performance Curves

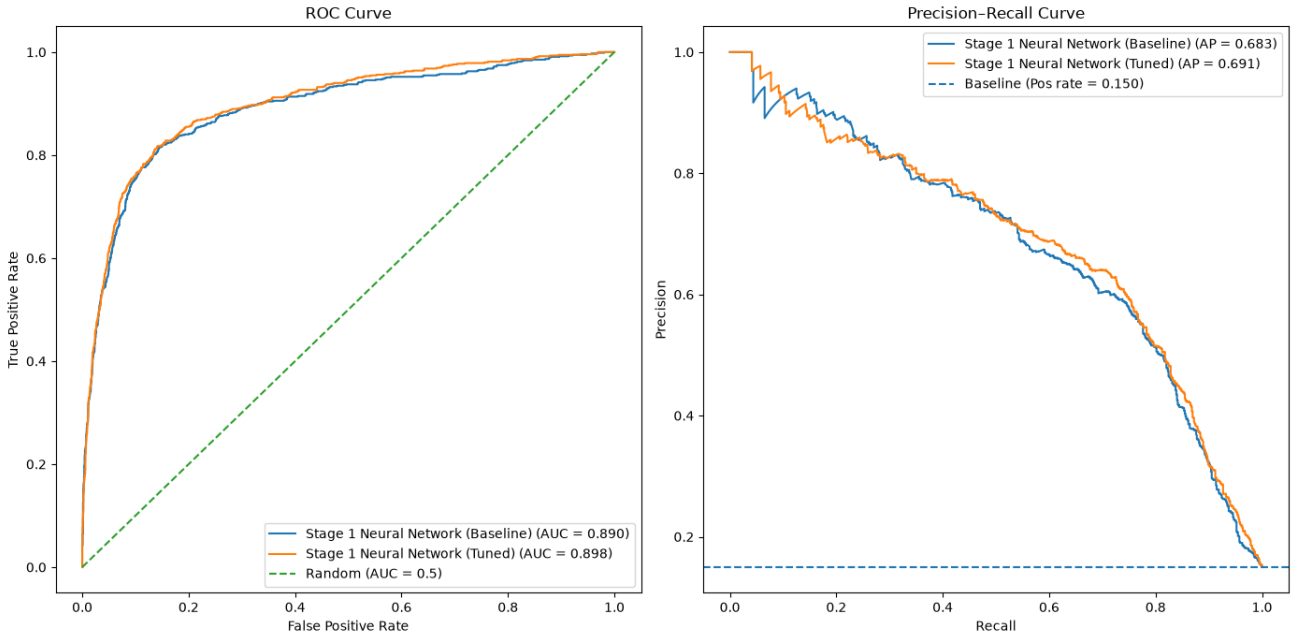


Figure G.1 – Neural Network Performance Curves – Stage 1

G.6 Stage 2 Neural Network Performance Curves

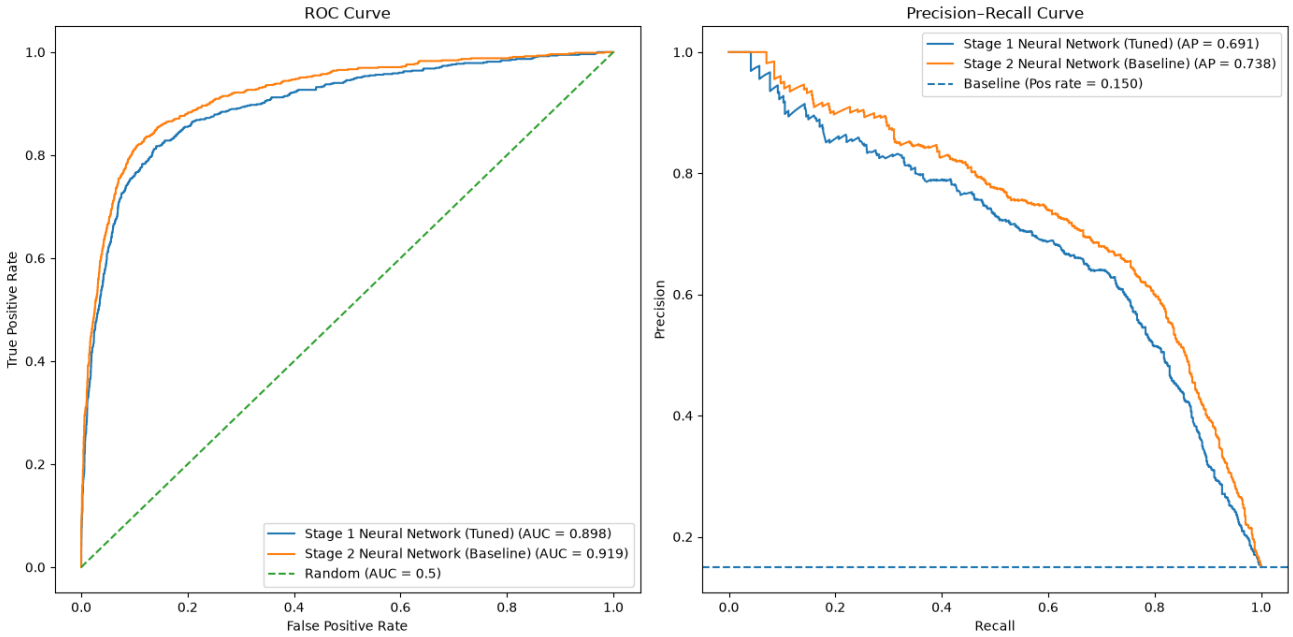


Figure G.2 – Neural Network Performance Curves – Stage 2 Baseline vs Stage 1

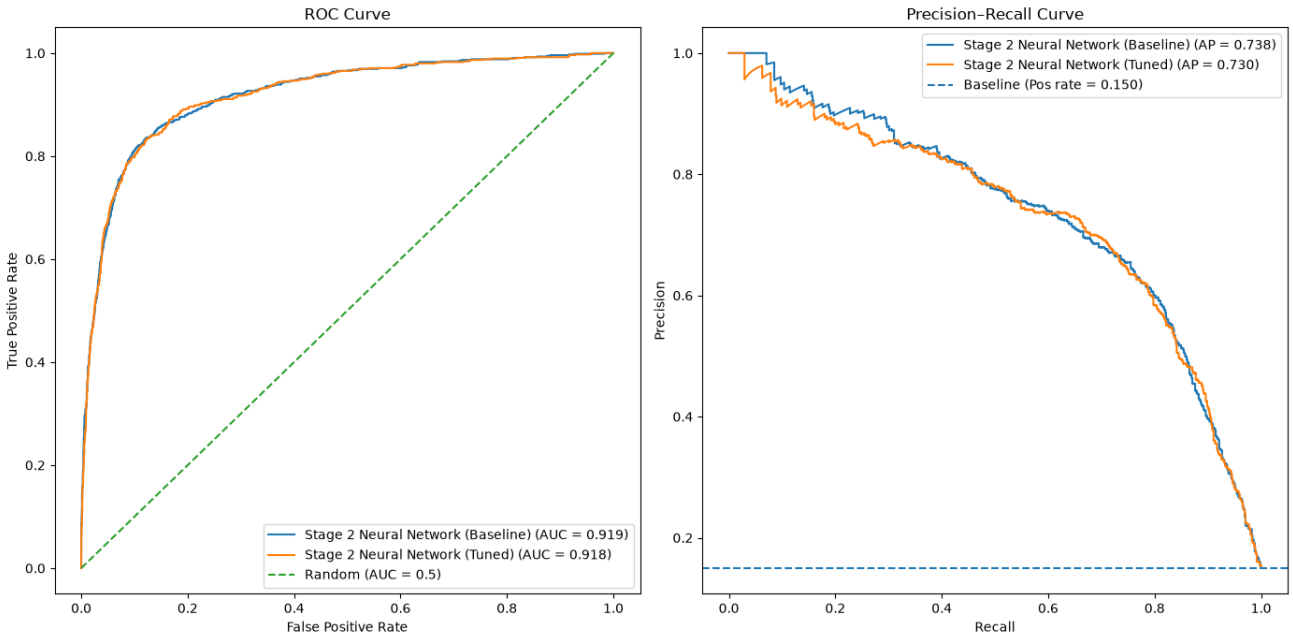


Figure G.3 – Neural Network Performance Curves – Stage 2

G.5 Stage 3 Neural Network Performance Curves

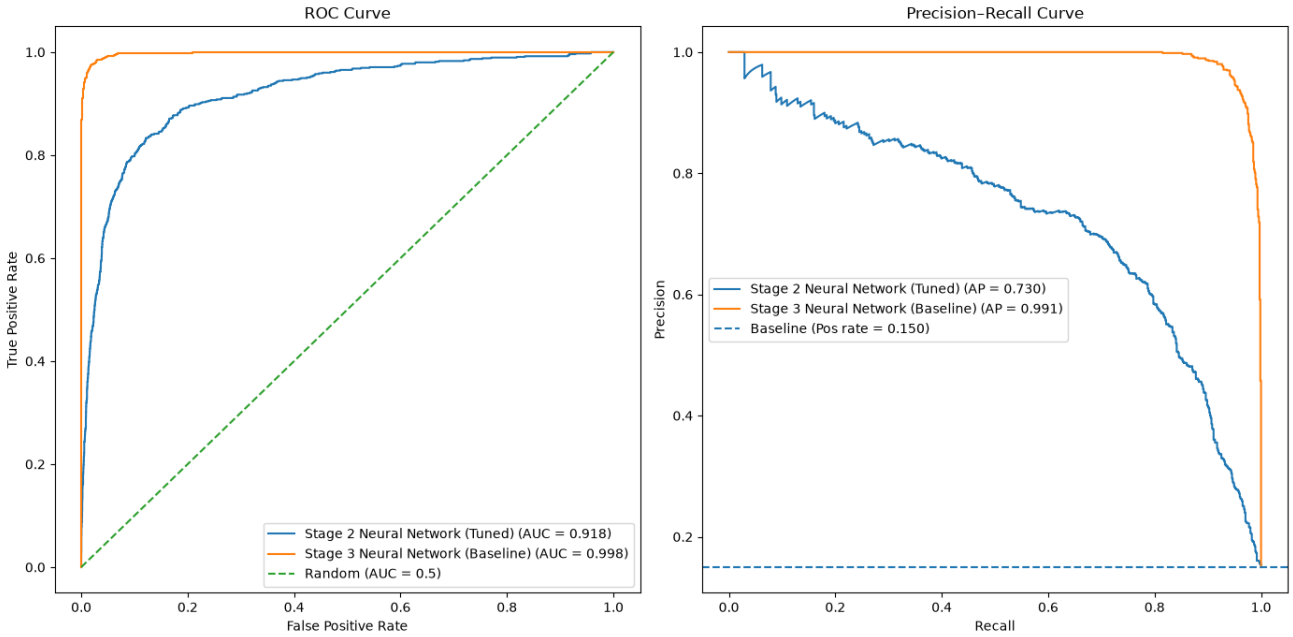


Figure G.4 – Neural Network Performance Curves – Stage 3 Baseline vs Stage 2

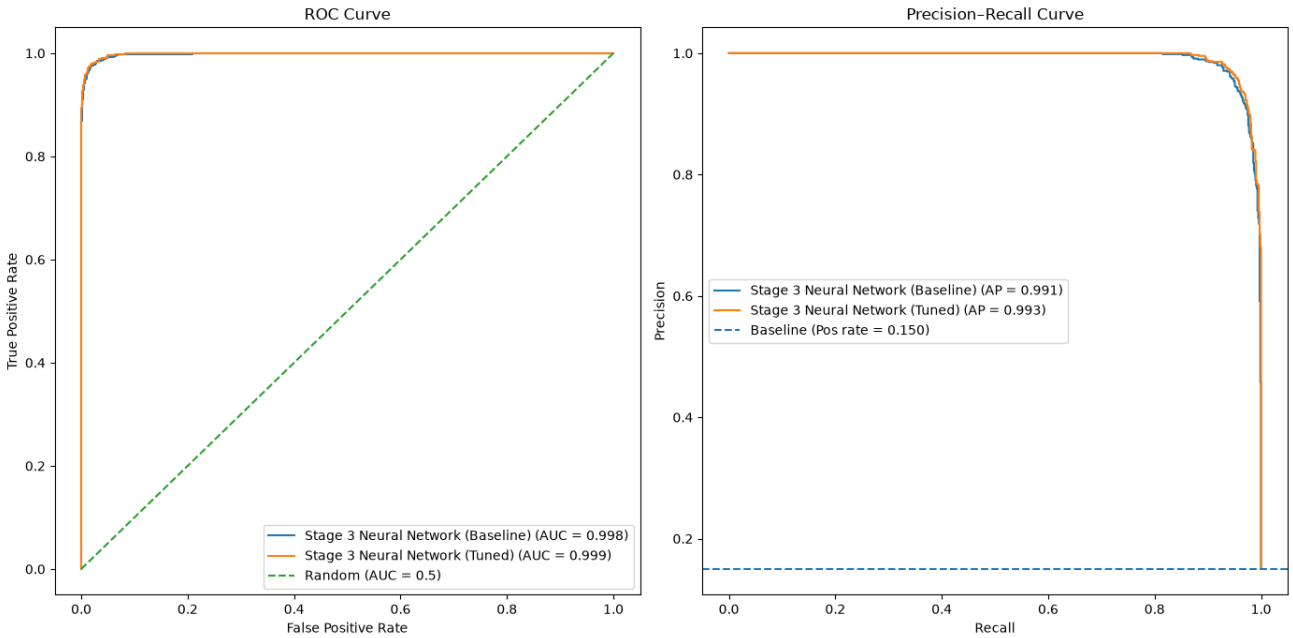


Figure G.4 – Neural Network Performance Curves – Stage 3